

Computer Vision (CV)

Polytechnic University of Tirana 18-29th May, 2026

CHRISTIAN MATA, PHD
christian.mata@upc.edu



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IMAGE PREPROCESSING

X					W		
0	0	0	0	0			
0	2	1	2	0	0.5	0.7	0.4
0	5	0	1	0	0.3	0.4	0.1
0	1	7	3	0	0.5	1	0.5
0	0	0	0	0			

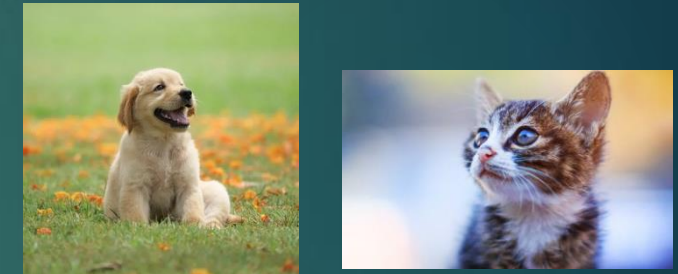
- Basic handling (math operations, reshape, logical masks)
- Filtering methods (Spatial, morphological, frequency)

IMAGE SEGMENTATION



Pixelclustering(unsupervised)
Pixel classification (supervised)
Deep Learning
(Encoder-Decoder)

IMAGE CLASSIFICATION



- Extraction of regional features
- Region-based features (Traditional ML)
- Deep Learning (CNNs)

Part 1

Module 1: Introduction to image processing (4 hours)

Basic image handling and preprocessing. Spatial (convolutional) filters, morphological operations, filters in the frequency domain.

Module 2: Image segmentation (4 hours)

Unsupervised segmentation using clustering algorithms. Supervised methods using pixelwise classifiers.

Module 3: Image classification (4 hours)

Extraction of regional features. Texture analysis and entropy. Region-based supervised classifiers. PCA Eigenfaces.

Part 2

Module 4: Introduction to deep learning (4 hours)

Convolutional neural networks for image classification. Segmentation using Convolutional Neural Networks.

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Computer Vision (CV)

Part 2: Schedule

TIRANA

	Monday 18	Tuesday 19	Wednesday 20		
9-10h	Module 1 Presentation Image Processing	Module 2 Image segmentation	Module 3 Image Classification		
10-11h					
11-12h					
12-13h					
13-14h					
14-15h					
15-16h					
16-17h					
17-18h					

ONLINE

	Monday 25	Tuesday 26	Wednesday 27	Thursday 22	Friday 23
9-10h					
10-11h					
11-12h					
12-13h					
13-14h					
14-15h					
15-16h	Test Activity	Working class No online class	Module 4 Artificial Intelligence		
16-17h					
17-18h					

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Course materials

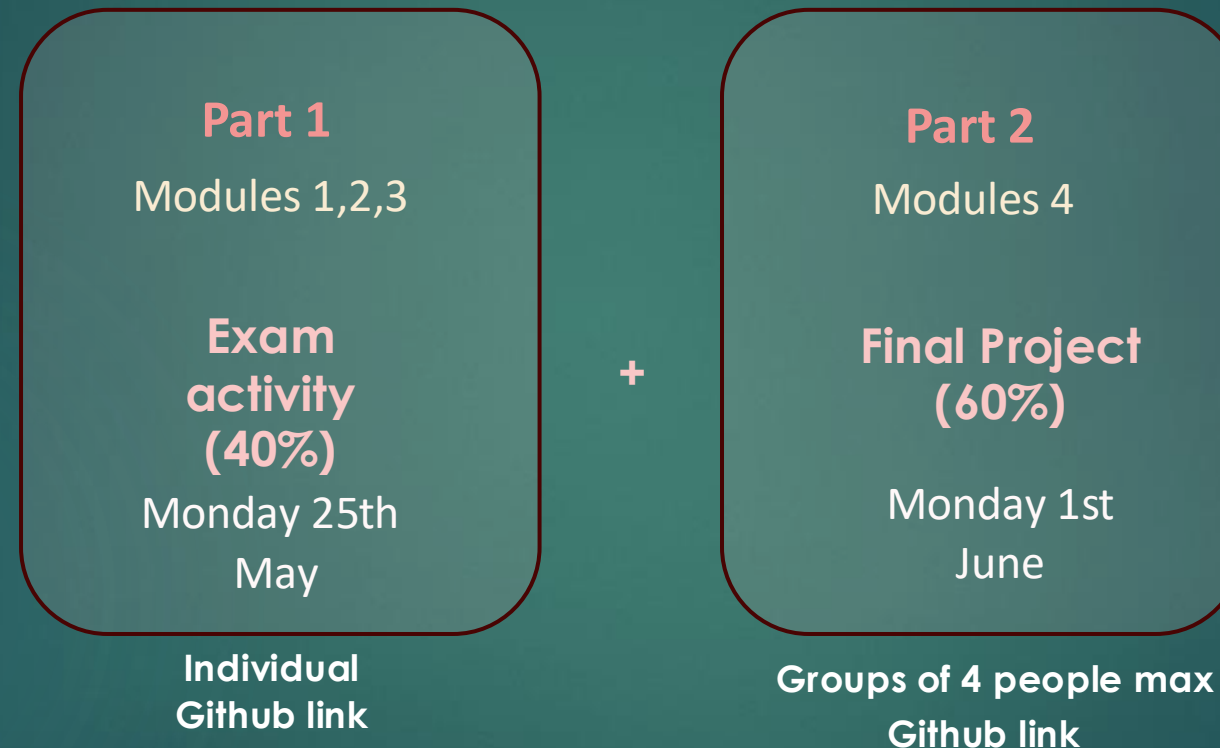


https://github.com/xtianu/CV_Tirana_2026



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► Activities & Evaluation



Via e-mail: christian.mata@upc.edu